High Helium Retentivity in Goethite under Hydrothermal Conditions Measured by Step-Heat Experiments in Silica Glass Ampoules

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Abstract

U-Th/He geochronology of goethite (FeOOH) dates weathering and ferruginization directly, but interpreting a goethite date requires knowing how the mineral retains radiogenic helium, which has proven difficult to measure. Recent work revealed that goethite decomposes to hematite and water during vacuum step heating, and helium release records dehydration kinetics rather than helium migration through stable goethite. We report the first helium step-heat experiment on goethite under hydrothermal conditions, made possible by sealing the sample and liquid water inside a pure silica glass ampoule, which retains water at up to 85 bar while transmitting helium to the vacuum system. Permeation through the ampoule wall delays measurement of released helium; we quantify this lag with a first-order permeation model and remove it either by measuring the ampoule's residual helium with low-temperature permeation steps or, for long steps, by bounding it. Encapsulated Durango apatite reproduces its published Arrhenius array under this correction, establishing that the ampoule contribution can be separated from the sample's. The ampoule's permeation rate constant is measured during the experiment itself from series of isothermal permeation steps, which is necessary because water-bearing ampoules permeate faster than published rates for anhydrous silica glass rather than more slowly; we attribute this to displacement of gas-phase volume by liquid water rather than to any change in the glass, and it makes published permeabilities a conservative lower bound.

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Proton-irradiated goethite Win-06-01b, previously step heated in vacuum, was held at water saturation over 170–250°C for 40 days through multiple prograde and retrograde cycles. Apparent helium diffusivities are approximately seven natural log units below those measured on the same sample in vacuum. Above 7% cumulative ^{3}He yield the array is linear, with an activation energy of 157.1 ± 5.3 kJ/mol, and its delta plot follows multiple diffusion domain systematics. A domain with these parameters is nearly quantitatively retentive of helium at 25°C over billion-year timescales. Because the difference between hydrothermal and vacuum experiments on the same aliquot exceeds the differences among vacuum experiments on unrelated samples, vacuum step heats of goethite appear to record the mineral's response to the experiment rather than helium migration through it, and helium retention in natural goethite is correspondingly more robust than those experiments imply. The method extends directly to systematic tests of crystallite size, chemical composition, and radiation damage.

Keywords: goethite, U-Th/He geochronology, helium diffusion, hydrothermal, silica glass permeation, step heating

1. Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). Goethite is a powerful chronometer for dating surface processes such weathering and ferruginization directly: laterite and duricrust formation and recrystallization (H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Allard et al., 2018), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. Step heats of proton-irradiated samples measure the degree of partial helium loss by the ^{4}He/^{3}He method (Shuster, P. M. Vasconcelos, et al., 2005), and reported losses are always non-zero but modest, ranging from 2 to 30% (Shuster, P. M. Vasconcelos, et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017), but the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved because goethite (FeOOH) decomposes to Hematite (Fe_2O_3) in the step heat

experiments used to characterize it (Farley, H. B. Monteiro, et al., 2024). Farley, H. B. Monteiro, et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Goethite dehydration is suppressed at high water pressure, so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. A particularly simple and agile approach to do this is to seal the sample inside a pure silica glass ampoule. Silica glass is the most helium-permeable common glass (Norton, 1953; Altemose, 1961) because it lacks the network-modifying cations that occupy interstitial space in other glasses. Helium permeation in silica glass is sufficiently rapid, so helium evolved from a sample inside the ampoule escapes to the vacuum system and can be measured while other gaseous species, such as water, sulfur, or oxygen is retained. Earlier work has exploited the permeability and selectivity of silica glass for U-Th/He geochronology (O. Yakubovich, Podolskaya, et al., 2020; O. Yakubovich, Vikentyev, et al., 2021; O. V. Yakubovich, Konstantinova, et al., 2024) and for monotonic ramped-heating experiments (O. V. Yakubovich, Gedz, et al., 2019), but not to impose hydrothermal conditions during step heating.

Silica glass encapsulation offers a unique opportunity to preserve reactive phases such as goethite that would otherwise decompose during helium step heat experiments Fig. 1, motivating the characterization of a step heat method utilizing this approach. Early applications reveal one cost: Permeation through the ampoule wall is rapid but not instantaneous, so helium released during a heating step is measured partly during that step and partly during later ones. We refer to this offset as permeation lag, and it must be quantified before any encapsulated step heat can be faithfully interpreted.

This paper establishes that the ampoule contribution to the measured helium in step heat experiments is tractable and that this method can address questions that vacuum experiments cannot. We show that encapsulation stabilizes reactive phases during heating and that we can successfully deconvolve the ampoule's permation lag effect from helium step heat measurements of Durango Apatite to recover its well-documented helium diffusion systematics (Wolf, Farley, and Kass, 1998; Farley, 2000). We show that the ampoule's permeation rate constant can be measured during the experiment

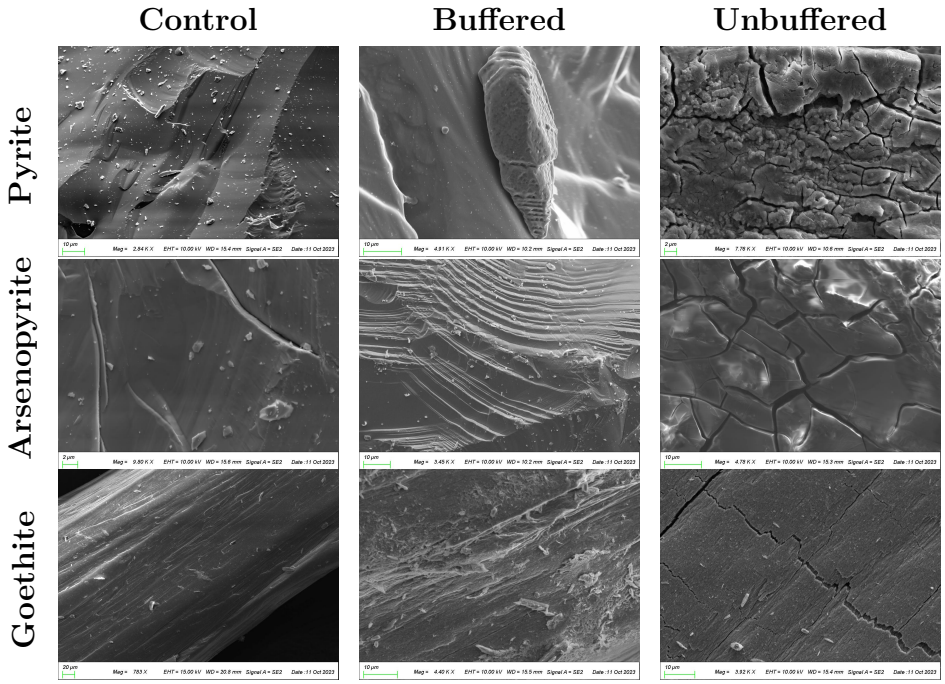


Figure 1: Secondary electron images of pyrite, arsenopyrite, and goethite before heat treatment (control), after heat treatment buffered within a sealed silica glass ampoule, and after heat treatment unbuffered. Sulfides were heated at 400°C for at least 4 hours, buffered with fused sulfur (pyrite) or mineral realgar (arsenopyrite); goethite was heated at 300°C, buffered with water for 4 hours and unbuffered in vacuum for 11 hours. Buffered samples retain the smooth surfaces and clear cleavage faces of the controls, with no evidence of decomposition. Unbuffered samples show contraction cracks and surface roughening. For goethite, cracks in the unbuffered material run both parallel and perpendicular to the long axis of the polycrystalline aggregate.

itself, which is necessary to apply the method to goethite because water-bearing ampoules often permeate heium anomalously fast. Finally, we apply the method to proton-irradiated goethite Win-06-01b under water saturation at 170–250°C, the first hydrothermal helium step heat reported for goethite.

2. Materials and Methods

2.1. Ampoule preparation and sealing

Segments of pure silica glass tube approximately 10 cm long, 4 mm outer diameter and 3 mm inner diameter, were sonicated in sequential baths of acetone, ethanol, and MilliQ water, then dried overnight at 100°C. The tubes were hand-pulled into pipettes with a jewelry welding torch and the thin ends were fused closed. The sample and buffering phase were then loaded into the pipette tip. Inner and outer diameters were measured on a calibrated stereo microscope. These give the geometric factor G of Eq. (4): 2393.5 cm^{-2} for the Durango apatite ampoule (850 µm inner, 1350 µm outer diameter) and, for the goethite ampoule, the value used throughout §4.2.

It is paramount to avoid heating the sample during the encapulation process to preserve the integrity of the step heat experimentl however, this challenging because silica glass softens well above 1600°C. To protect against heating during the sealing process, the sample-end of the ampoule was held by a water-soaked Kimwipe while the top was fused. Liquid water saturated the wipe afterward, so the adjacent glass containing the sample never exceeded 100°C. For goethite, the contents were frozen and held in liquid nitogen instead, serving the same purpose while additionally preventing the loss of the water charge.

Before sealing, each ampoule was pumped to below 0.1 torr through its open top. The purpose of evacuation is mechanical: under an internal-external pressure contrast, softening glass deforms inward and self-seals, which gives a better seal with far less heat input. Unfortunately, this process may not usefully reduce the ampoule's helium inventory, because room-temperature permeation is fast enough that atmospheric helium re-enters a sealed ampoule within hours to days. The residual atmospheric helium is instead measured directly at the start of each experiment (§3.3).

Ampoules containing water withstood the vapor pressure of liquid-water saturation up to about 86 bar at 300°C (Haar et al., 1984) but ruptured above 315°C. We recommend limiting water-bearing ampoules to below 300°C.

2.2. Encapsulated Durango apatite

Durango apatite crushate sieved to 300 µm (radii ≈ 150 µm), 3.18 mg, was encapsulated (Fig. 2a) and step heated over 100–400°C. Durango apatite is a standard for helium diffusion and U-Th/He thermochronometry (Zeitler et al., 1987; Wolf, Farley, and Silver, 1996; Farley, 2000; Shuster and Farley, 2005; Shuster, Flowers, et al., 2006; Flowers, Shuster, et al., 2007), so its Arrhenius behavior is known independently. The experiment then is a direct test of the validity of the permeation-lag correction.

Each high-temperature step was preceded and followed by a sequence of two or three low-temperature permeation steps (§3.3). The total ^{4}He yield was estimated from the measured mass and the helium concentration of 194.2 nccSTP/mg reported by Farley (2000), a value verified independently in-lab. The atmospheric ^{4}He present at the start of the experiment, constrained by the initial permeation steps, was 1.47 ± 0.18 nccSTP. The full step-heat dataset is given in the Supplementary Information (Table **??**).

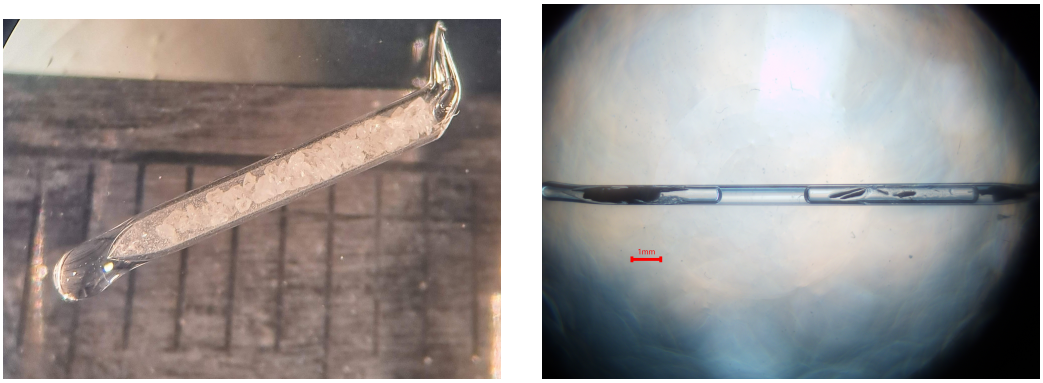
2.3. Hydrothermal goethite experiment

The sample is dense colloform goethite Win-06-01b, collected as float in the Flinders Ranges, South Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages, 180–135 Ma (Waltenberg, 2012). Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following Shuster and Farley (2005) to produce a uniform ^{3}He distribution. Aliquots of this same irradiated material were previously step heated in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024), whose results agree well with one another.

An aliquot of 1.18 mg Win-06-01 goethite blades shorter than 2 mm was transferred to the ampoule tip in the presence of an ionizing fan. Less than one microlitre of distilled water was added through a glass capillary, submerging the goethite. The ampoule was tapped to release bubbles before it was flash-frozen in liquid nitrogen, the it was transferred to a −10°C saltwater-ice bath so that the water ice remained solid while atmospheric gases were pumped away. After pumping to <0.1 torr, the saltwater bath was swapped with liquid nitrogen to prevent heating of the sample or melting of the water ice buffer, and the ampoule was fused closed.

The sealed ampoule, designated HT-Win-06-01b, retained all of its original water through the entire experiment (Fig. 2b), across a pressure contrast of more than eleven orders of magnitude between an internal vapor pressure

of roughly 40 bar at 250°C (Haar et al., 1984) and the high vacuum (10^{-7} torr) pressures of the lamp cell.



(a) (b)

Figure 2: Silica glass ampoules used for helium step heat experiments. (a) A typical ampoule as prepared, here containing Durango apatite crushate sieved to 300 µm (radii ≈ 150 µm). The ampoule is roughly one cm long and one mm wide; the bars visible behind it are separated by 1 mm. (b) HT-Win-06-01b after completion of the step heats, before breaking for the final furnace step. At room temperature this ampoule is about half gas, one third liquid water, and one sixth goethite shards under 2 mm long. All of the original water remains, across a pressure contrast sustained for over a week at 250°C. The gel-like lining on the inner wall is likely amorphous silica dissolved from the ampoule and redeposited by the thermally cycled hydrothermal fluid.

Before heating, HT-Win-06-01b was held at 70°C for four days under vacuum to remove atmospheric helium entrained in the ampoule. The experiment then ran for about 40 days over 170–250°C, with step durations from 2 hours to about a week, through multiple prograde and retrograde cycles designed to test for multiple diffusion domain behavior. We did not heat above 250°C because of the risk of rupture due to high vapor pressure.

Temperature records show oscillations of about 1°C amplitude around each set point. While we did not conduct a detailed study to charectarize temperature gradients throughout the ampoule during step heats, we suggest that the apatite experiment of §2.2 provides an independent check that recorded temperatures are accurate.

For the final step, the ampoule was broken and dried overnight at 50°C, then helium was completely degassed at ∼1100°C in a resistance furnace and measured on a Thermo Fisher Helix SFT mass spectrometer.

2.4. Heating and helium isotope analysis

Sealed ampoules were washed in ethanol and air dried, then placed in copper foil packets and mounted to a K-type thermocouple in an ultra-high-vacuum projector-lamp heating cell (Farley, Reiners, et al., 1999). The lamp cell was pumped to high vacuum and held at 100°C for 30 minutes to desorb adsorbed species. For encapsulated Durango apatite, less than 0.5 ‰ of the radiogenic helium would be expected to escape during this pre-treatment, about 0.01 nccSTP, using the diffusion parameters of Farley (2000) and the forward equations of Fechtig and Kalbitzer (1966).

Helium isotope abundances for the Win-06-01b goethite were measured on a Thermo Fisher Helix-MC-Plus^{10K}, and 4He abundances were measured with isotope dilution on a Pfeiffer QMS 200 mass spectrometer, both at Caltech. Evolved gases were purified over a liquid-nitrogen charcoal trap, a titanium sponge getter, and a SAES NP10 getter. For the Win-06-01b step heats, ^{3}He was counted on a pulse-counting electron multiplier and ^{4}He on a Faraday detector. Sensitivities were tracked with measurements of calibrated gas standards every 8 hours, giving internal reproducibilities of 5% for ^{3}He and 2% for ^{4}He (2σ).

For HT-Win-06-01b, a cold blank gave a negligible ^{3}He blank and a ^{4}He blank of about 1.25 pccSTP, comparable to the 4He signals measured during low-temperature permeation steps, but 3He was well-above blank for these steps. Pump-out between steps was 1–5 minutes.

2.5. Electron microscopy of thermally treated phases

To assess whether encapsulation preserves reactive phases at experimental temperatures, single crystals of pyrite and arsenopyrite and aggregates of Bahia goethite morphologically similar to Win-06-01b (Shuster, P. M. Vasconcelos, et al., 2005; H. Monteiro, 2017) were subjected to heat treatment in both buffered and unbuffered configurations: unbuffered samples were heated either in vacuum in the lamp-heating apparatus of Farley, Reiners, et al. (1999) or in open air while buffered samples were sealed in silica glass ampoules with fused sulfur, mineral realgar, or water for pyrite, arsenopyrite, and goethite respectively. Sulfides were held at 400°C for at least 4 hours and goethite at 300°C for 4–11 hours. The ampoules were then broken and the samples air dried, then mounted on carbon tape, carbon coated, and imaged in secondary electron mode on a ZEISS 1550VP field emission SEM at Caltech.

3. Permeation Systematics of Fused Silica Ampoules

3.1. Helium budget of a single step

Consider one heating step. Helium enters the ampoule interior from the sample and leaves it by permeation through the wall, so

$$n_t = n_0 + \omega - m. \tag{1}$$

Here n_0 is the helium in the ampoule at the start of the step, n_t is the helium remaining at its end, ω is the helium released by the sample during the step, and m is the helium that permeated out and was measured. Only m is observed; but the quantity of interest is ω.

Steps are consecutive, so the helium inventory quantified between steps applies to both the previous and the subsequent step:

$$n_t^i = n_0^{i+1}, \qquad n_0^i = n_t^{i-1}. \tag{2}$$

Recovering ω for a step requires knowledge of the ampoule's helium content immediately before and immediately after it. Sections 3.3 and 3.5 give the two routes we use to obtain those quantities.

When ω is large compared with n_0 and n_t, Eq. (1) reduces to $\omega \approx m$ and no correction is needed. That condition does not hold for hour-long steps at 100–400°C, such as those used on the Durango apatite, but it is approached for long steps used with retentive phases, such as Win-06-01b (§3.5).

3.2. Assumptions and the permeation rate law

Helium permeation through silica glass is a well-understood first order process (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968). Two conditions must hold before permeation can be described by a first-order rate law.

1. The helium partial pressure outside the ampoule is negligible compared with that inside. The ampoule and diffusion cell volumes differ by a factor of roughly 10,000 (Farley, Reiners, et al., 1999), so this holds throughout.
2. The helium concentration profile across the glass wall is at diffusive steady state. This is established within 30–70 minutes at 80°C for the ampoules used here, and it is disturbed whenever the temperature changes, because the glass/gas partitioning of helium is temperature

dependent. We therefore hold the ampoule at the permeation temperature for at least one diffusive timescale $\tau = L^2/D$ after every temperature change before measuring, where L is the wall thickness and D the diffusivity of helium in silica glass (Swets et al., 1961).

Given these conditions, permeation is first order in the amount of helium present:

$$\frac{dn}{dt} = -GKnRT, \tag{3}$$

where n is the number of helium atoms, G is a geometric factor with units of cm^{-2}, K is the permeation constant for helium in fused silica, which follows an Arrhenius law $K(T) = K_0 e^{-E_a/RT}$ and has been measured repeatedly (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), R is the gas constant, and T is temperature in Kelvin. Equation (3) is the standard permeation equation (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), written in terms of atom counts rather than partial pressures via the ideal gas law and taking the external partial pressure as zero.

For a cylindrical wall with negligible permeation through the ends, the geometric factor is

$$G = \frac{2}{a^2 \ln(b/a)}, \tag{4}$$

with a the internal and b the external radius (Masaryk, 1968). There is no dependence on length. G varies with ampoule construction and must be determined for each experiment.

3.3. Recovering the residual helium with permeation steps

At temperatures low enough that the sample releases no helium, $\omega = 0$ and the ampoule's helium inventory changes only by permeation. The measured helium from such a step then is a measurement of the helium inventory of the ampoule. To this end, a useful expression for the fractional yield of an isothermal step is derived, first through integration of Eq. (3) by separation of variables, yielding $n_t = n_0 e^{-GKRTt}$, then by relating the measured value to the tube contents by Eq. (1) ($m(\omega = 0) = n_0 - n_t$).

$$^{He}F = \frac{m}{n_0} = 1 - e^{-\lambda t}, \qquad \lambda = KGRT, \tag{5}$$

This equation is applied to recover the helium inventory at the beginning of the step.

$$n_0 = \frac{m}{{}^{He}F}. \tag{6}$$

where ${}^{He}F$ is the fraction of the ampoule's helium inventory that permeates out during a step of duration t, and λ is the ampoule's permeation rate constant. Collecting K, G, R, and T into λ is useful because λ is the quantity that can be measured directly during an experiment (§3.4), whereas K and G cannot be separated without independent knowledge of the ampoule geometry.

We call such a low-temperature step a *permeation step*. A permeation step does double duty: through Eq. (2) it constrains both the helium remaining after the preceding high-temperature step and the helium inherited by the following one. Placing a permeation step after every high-temperature step therefore supplies n_t and n_0 throughout the experiment, and Eq. (1) returns ω. This is the approach used to reduce the encapsulated Durango apatite step heat data(§4.1).

3.4. Measuring the ampoule's rate constant during the experiment

Applying Eq. (6) requires λ. Published permeabilities for anhydrous silica glass (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968) combined with a measured G give an estimate, but they should not be relied upon, and they cannot be relied upon at all in hydrothermal experiments: water entering the glass changes its permeability (Shelby, 1972; Tsuru et al., 2011), and water-bearing ampoules show anomalously high rate constants (§5.3). The rate constant should therefore be measured on the ampoule actually in use.

Because permeation remains first order regardless of the values of K and G, a series of consecutive isothermal permeation steps determines λ and n_0 together with a non-linear least squares solve.

$$m_{i+1} = n_0 \left(1 - e^{-\lambda t_{i+1}}\right) - n_0 \left(1 - e^{-\lambda t_i}\right). \tag{7}$$

We fit individual step yields rather than cumulative yields, because cumulative fitting propagates the errors of early measurements into later measurements, unduly biasing the solver to fit earlier measurements. We used the Levenberg-Marquardt implementation in SciPy (Virtanen et al., 2020), weighted by each measurement's own uncertainty, with initial guesses $\lambda \approx$

$6 \times 10^{-6}\,\mathrm{s}^{-1}$ and $n_0 \approx m_i/(1 - e^{-\lambda t_i})$ evaluated at the first measurement of each series. Uncertainties come from the fit covariance matrices. The fit is poorly conditioned unless enough time has elapsed for the cumulative yield to show curvature, which in practice means letting the permeation rate slow noticeably across three or more consecutive measurements. The exercise can technically be done with as few as two consecutive measurements, although the solver would be unable to estimate uncertainties on the fit parameters.

3.5. Bounding the residual helium for long steps

Permeation steps require a measurable signal at low temperature. When the sample is retentive and its yields are small, the low-temperature signal can fall to blank level and Eq. (6) cannot be applied. This is the situation for ^{4}He in the goethite experiment (§4.2). An alternative apporoach is available when steps are long compared with the permeation timescale of the ampoule, which is common for retentive phases.

During a sufficiently long step the ampoule approaches a steady state in which the helium outflux of permeation is balanced by release from the sample, so the residual inventory stops growing while measurable helium continues to accumulate. As time proceeds after this condition is met, the helium inventory becomes relatively small compared to the measured abundance and $\omega \approx m$ becomes an increasinly accurate approximation. Additionaly, an upper bound constraint on the ampoule's helium inventory can be calculated from the high temperature experimental measurement alone. The residual is then bounded above by

$$n^{ss} \leq \frac{m}{\lambda t - 1}, \tag{8}$$

which holds once λt is large enough for steady state to be assumed. When Eq. (8) is applied to a long step and to the step preceding it, Eq. (1), can be used to bracket ω, the true abundance of helium evolved from the sample:

$$m_1 - \frac{m_0}{(\lambda t)_0 - 1} \leq \omega_1 \leq m_1 \left(1 + \frac{1}{(\lambda t)_1 - 1}\right). \tag{9}$$

Equation (8) assumes the sample's release rate is constant at a given temperature. In practice the release rate declines with time for volume diffusion (Wolf, Farley, and Silver, 1996; Reiners and Farley, 1999; Farley and McKeon, 2015; Farley, 2018; Farley, H. B. Monteiro, et al., 2024) and for first-order processes (O. V. Yakubovich, Shukolyukov, et al., 2010; Wilson

and Bisson, 1973), which makes the true residual smaller than the bound and Eq. (8) correspondingly conservative. The exceptions are initially inhomogeneous helium distributions at low cumulative yields (Goldsmith et al., 2020; Ketcham et al., 2011; Shuster, Farley, et al., 2004; Farley, Wolf, et al., 1996) and radiation damage annealing (Cherniak, 2019; Guenthner et al., 2013; Flowers, Shuster, et al., 2007; Flowers, Ketcham, et al., 2009; Shuster, Flowers, et al., 2006), which can cause can cause helium release to accelerate. Additionally, equation (8) is conservative when the value of λ used does not exceed the true one, because a lower bound on λ yields an upper bound on the residual. Any λ that can be independently shown to underestimate the ampoule's true rate constant is therefore admissible. This bounding appoach is an excellent fallback when small yields during permeation steps of retentive phases cannot be well-resolved.

4. Results

4.1. Encapsulated Durango apatite step heats

Without correction, apparent diffusivities from the encapsulated Durango apatite step heat are irregular and would not be recognized as belonging to this well-characterized sample (Fig. 3). Applying the permeation-step correction of §3.3 removes this irregularity. The corrected diffusivities agree, within uncertainty, with the Arrhenius relationship reported by Farley (2000) for Durango apatite of the same 150 µm half-prism width, except for the first two plotted points and the 400°C step (Fig. 3), discussed below in §5.2. Four of the 25 high-temperature steps had permeation lag corrected yields within 2σ of zero and are excluded from the figure.

4.2. Rate constant of the hydrothermal goethite ampoule

^{3}He measurements from ten series of three to four consecutive isothermal permeation steps, held at 90–110°C for 120–180 minutes each with 26 minutes of cooling before inlet, were fit to Eq. (7) (Fig. 4). Fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration. One further series that otherwise qualified was poorly conditioned and was excluded; however, its fitted λ agreed with the others within uncertainty.

Converting these rate constants to permeabilities through $K = \lambda/GRT$ and regressing $\ln K$ against inverse temperature over 90–110°C gives $R^2 =$

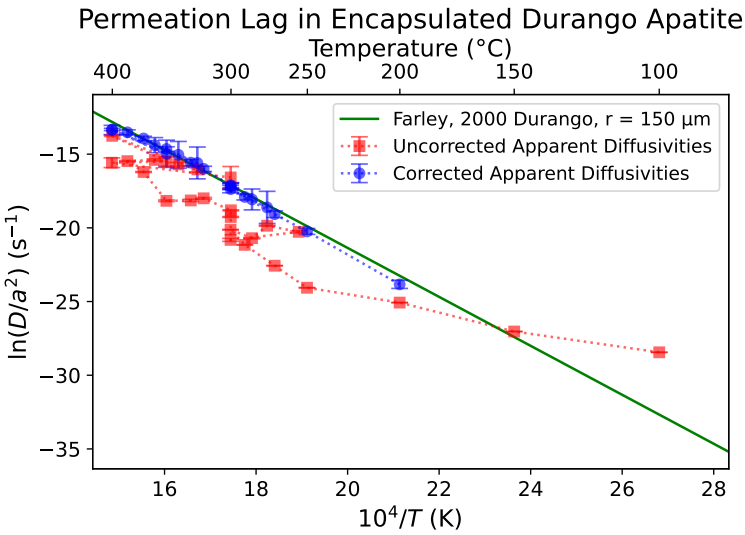


Figure 3: Arrhenius plot of a helium step heat of silica glass encapsulated Durango apatite, with 2σ uncertainties. Without permeation lag corrections the apparent diffusivities are highly irregular. With corrections the sample agrees with the diffusion parameters reported by Farley (2000) for the appropriate sieve size ($r = 150\,\mu m$). Beyond 300°C the activation energy decreases to give a slight downward curl, likely caused by annealing of crystal defects and radiation damage (Wolf, Farley, and Silver, 1996; Farley, 2000). Apparent diffusivities are calculated with the equations of Fechtig and Kalbitzer (1966). Four corrected diffusivities are within 2σ of zero and are excluded. The full dataset is given in Table **??**.

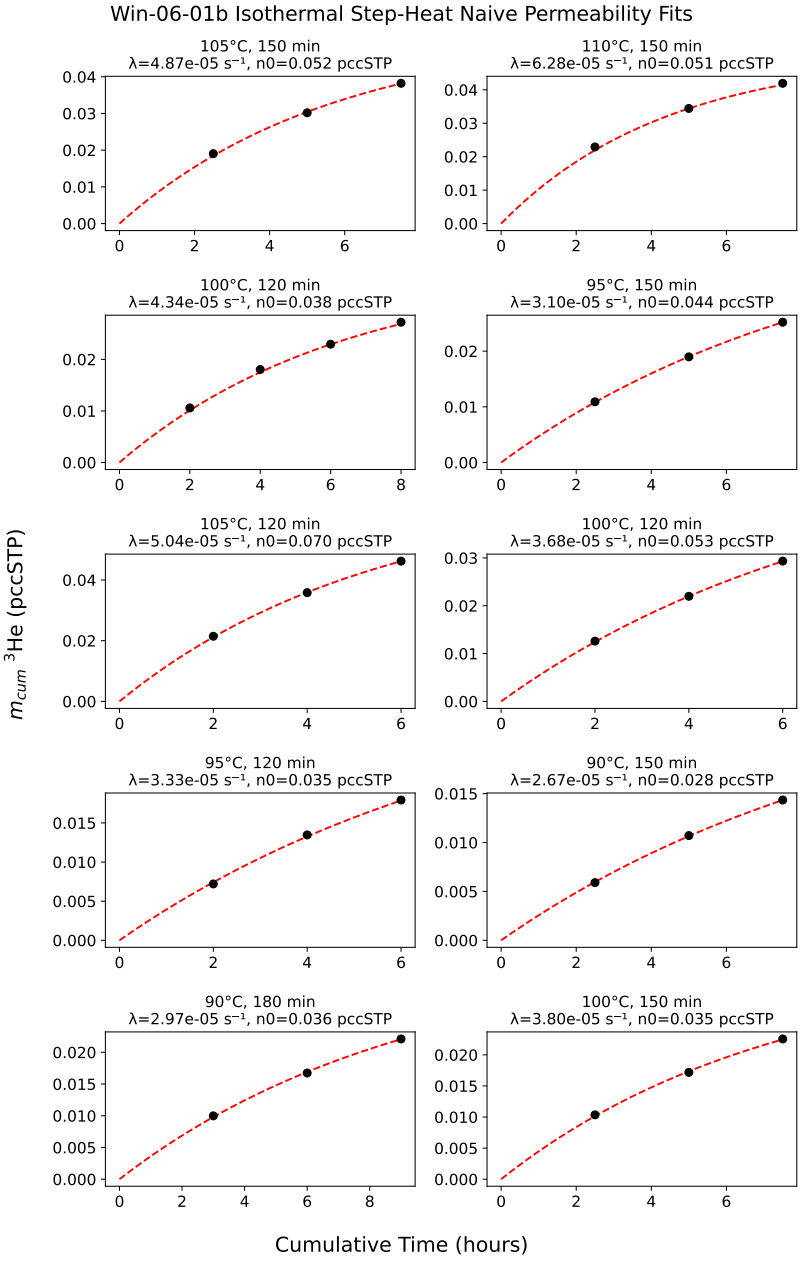


Figure 4: Cumulative ^{3}He yield against cumulative time for each series of consecutive isothermal permeation steps on HT-Win-06-01b, with the non-linear least squares fit of λ and n_0 to Eq. (7) overlain. Each series is held at a single temperature between 90 and 110°C. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly (§3.4). The rate constants recovered here are converted to permeabilities, $K = \lambda/GRT$, and plotted against temperature in Fig. 5.

0.936 across 10 series, with an activation energy of 10192 ± 1882 cal/mol (2 s.e.) (Fig. 5).

A finite-difference cooling simulation of permeation during the ampoule's cooling before measurement predicts that measurements will slightly overestimate the apparent permeability by an offset that is near constant across the temperature range. This offset is smaller than the 2σ uncertainties on the fitted values (Fig. 5). The methods for this simulation are described in the supplement.

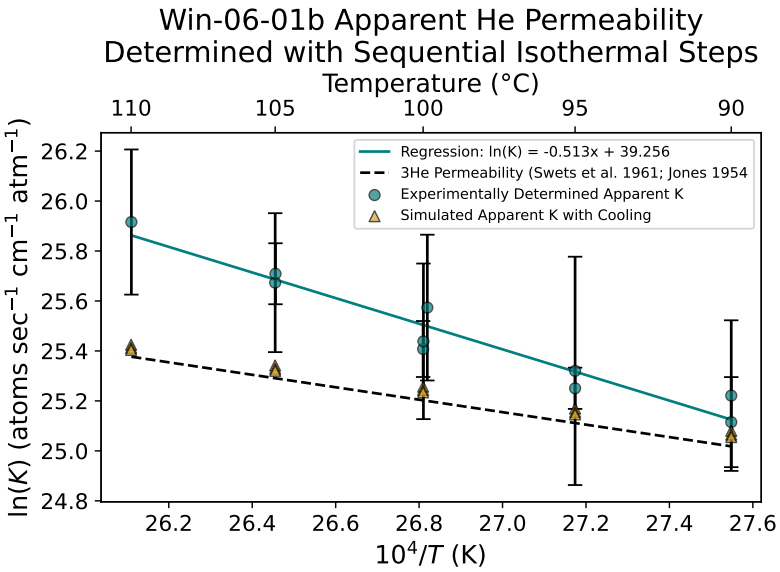


Figure 5: Apparent permeation coefficients for HT-Win-06-01b against inverse temperature, converted from the rate constants fitted in Fig. 4 via $K = \lambda/GRT$. The activation energy is higher than for anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). The apparent ^{3}He permeability exceeding the literature values of Swets et al. (1961) and Jones (1953) is not expected, and is attributed to displacement of gas-phase volume by liquid water (§5.3). Points labelled "Diffusion Model with cooling, Anhydrous" give the excess apparent permeability predicted for a dry ampoule that continued to permeate during the 26 minute cooldown before measurement, simulated over the same 120–180 minute holds at 90–110°C. Their scatter reflects differences in hold duration and is small; the offset from the literature values is near constant and much smaller than the uncertainty on the fitted values.

Table 1: ^{3}He step-heat dataset for HT-Win-06-01b under the "treatment as uncertainty" method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see §3.4). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 9 and others derived from this dataset).

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.158	-21.775	-21.503
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.715	-21.406	-21.160
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.345	-21.068	-20.843
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.823	-20.604	-20.392
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.516	-21.246	-21.148
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.836	-22.124	-21.900
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.479*	-22.863*	-22.646*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.053*	-23.731*	-23.618*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.839*	-24.404*	-24.207*
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.168*	-24.847*	-24.665*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.534*	-23.360*	-22.812*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.984*	-22.758*	-22.554*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.587	-21.553	-21.475
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.948	-20.811	-20.579
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.062	-19.913	-19.667
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.176	-19.778	-19.474
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.626	-19.296	-18.942
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.801	-19.874	-19.601
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.176	-20.487	-20.250

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.284	-20.864	-20.650
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.885	-21.421	-21.224
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.009	-21.781	-21.650
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.605	-21.462	-21.299
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.258	-21.107	-20.936
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.913	-20.750	-20.566
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.495	-20.326	-20.125
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.181	-19.973	-19.743
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.261	-19.959	-19.734
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.285	-19.979	-19.755
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.989	-19.976	-19.969
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.026	-20.778	-20.689
Total			146.1483						

Table 2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the "treatment as uncertainty" method, structured as in Table 1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 9. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 1.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.207	-25.965	-25.619	–	197.3
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.314	-25.022	-24.717	248.2	253.0
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.407	-24.152	-23.876	326.3	330.1
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.425	-23.213	-22.956	403.7	410.6
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.807	-23.527	-23.413	442.6	463.6
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.945	-24.225	-23.976	429.1	505.9
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.661*	-24.942*	-24.701*	392.4*	504.2*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.165*	-25.796*	-25.675*	446.7*	500.9*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.586*	-26.808*	-26.592*	224.7*	350.1*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.475*	-26.340*	-26.141*	843.2*	843.2*
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.787*	-25.411*	-24.791*	375.6*	475.5*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.067*	-23.021*	-22.786*	2667.1*	2667.1*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.499	-23.302	-23.214	471.7	552.7
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.391	-22.249	-21.993	575.1	590.8
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.360	-21.203	-20.931	612.1	633.4
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.289	-20.899	-20.556	665.9	718.7
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.906	-20.469	-20.072	541.7	640.2
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.043	-20.981	-20.680	469.1	670.4
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.376	-21.575	-21.315	505.5	674.3

continued

(continued)

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.386	-21.922	-21.688	599.8	689.6
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.956	-22.448	-22.232	608.3	707.9
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.136	-22.863	-22.721	608.2	666.5
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.679	-22.523	-22.345	652.1	678.1
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.290	-22.128	-21.940	674.6	701.7
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.892	-21.719	-21.518	705.1	734.6
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.479	-21.290	-21.070	694.4	731.9
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.134	-20.906	-20.659	703.3	747.3
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.250	-20.908	-20.661	653.7	729.2
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.282	-20.932	-20.647	646.8	721.0
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.704	-20.691	-20.682	800.6	802.1
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.688	-21.422	-21.325	770.5	819.4
Total			145.8145								997.7

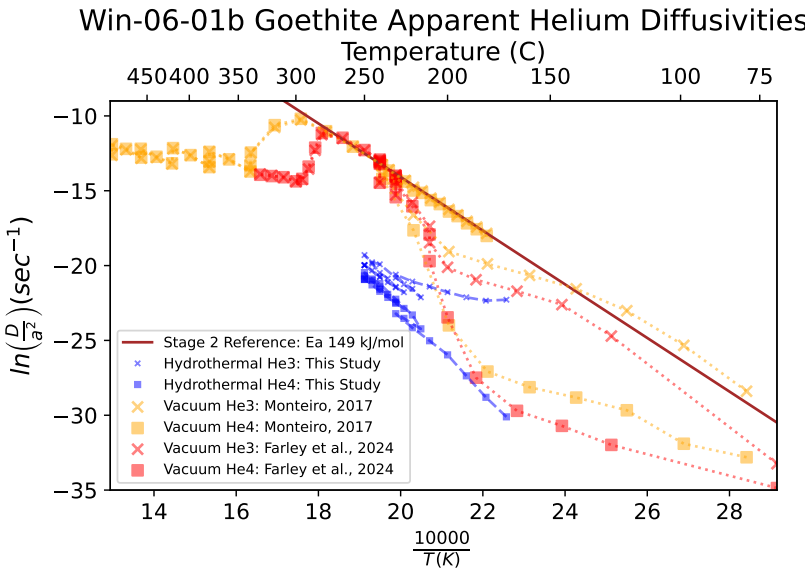


Figure 6: Arrhenius plot of ^{3}He and ^{4}He apparent diffusivities from HT-Win-06-01b, with step heats of the same Win-06-01b material in vacuum from H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). The Stage 2 reference line is fit to the step heats of H. Monteiro (2017). Error bars on the data from this study are omitted for clarity. A diffusion domain corresponding to the Stage 2 reference line would have a closure temperature of about 45°C at a cooling rate of 10°C/Ma.

The bounding method (§3.5) was used to reduce the Win-06-01b step heat data (see §5.4 for details). The first three experimental steps have $\lambda t < 4$ and no bound could be evaluated; they are excluded from further analysis. Later steps with $\lambda t < 4$ could be treated, because bounds on the preceding steps' residuals were available. Tables 1 and 2 give the resulting dataset.The ampoule retained its full water charge for the duration of the experiment (Fig. 2b), so hydrothermal conditions were maintained throughout.

Apparent diffusivities for HT-Win-06-01b are lower than those measured for the same sample in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) at every step. This gap widens to approximately seven natural log units after the first few steps (Figs. 6, 7).

The ^{3}He Arrhenius array sharply declines along a concave-up trend to about 3.5% cumulative ^{3}He yield, then decreases slowly along sub-parallel

arrays through the prograde and retrograde cycles between 3.5 and 7% cumulative yield. Beyond about 7% cumulative yield the array is linear. A regression over the remaining 12 steps, spanning 7–15% cumulative ^{3}He yield, gives $R^2 = 0.989$ and an activation energy of 157.1 ± 5.3 kJ/mol. Treating this array as a single diffusion domain and considering only the uncertainty in the intercept gives a closure temperature of 98.1 ± 8.6 °C at a cooling rate of 10°C/Ma. Total ^{3}He released over the experiment was 14.7% of the sample's total ^{3}helium inventory.

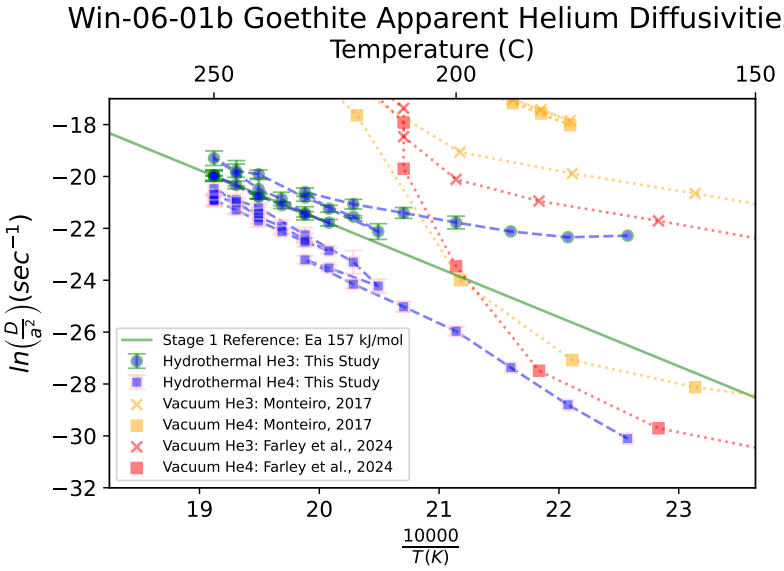


Figure 7: Linear regression of the ^{3}He apparent diffusivities of HT-Win-06-01b above 7% cumulative yield, defining the Stage 1 reference line. A diffusion domain corresponding to this line has a closure temperature of 98.1 ± 8.6 °C at a cooling rate of 10°C/Ma. Apparent diffusivities calculated from ^{4}He are sometimes within error of those calculated from ^{3}He, but are occasionally resolved at the 2σ level and lie about one log unit lower after a cumulative ^{3}He yield of 14%.

The ^{4}He array differs in shape. It is concave-down to about 0.1% cumulative ^{4}He yield, after which it follows a roughly linear array about one log unit below the ^{3}He array.

On a delta plot constructed against the ^{3}He regression as reference line, ^{3}He delta values decrease monotonically until about 7% cumulative loss and are within error of the reference line thereafter (Fig. 8). ^{4}He delta values

are more uncertain; they increase initially and settle into a slow increase or plateau about one log unit below the ^{3}He reference line by about 5% cumulative ^{3}He yield. A group of low-yield measurements on the first retrograde cycle produce scattered, high-uncertainty delta values just after 4% cumulative loss.

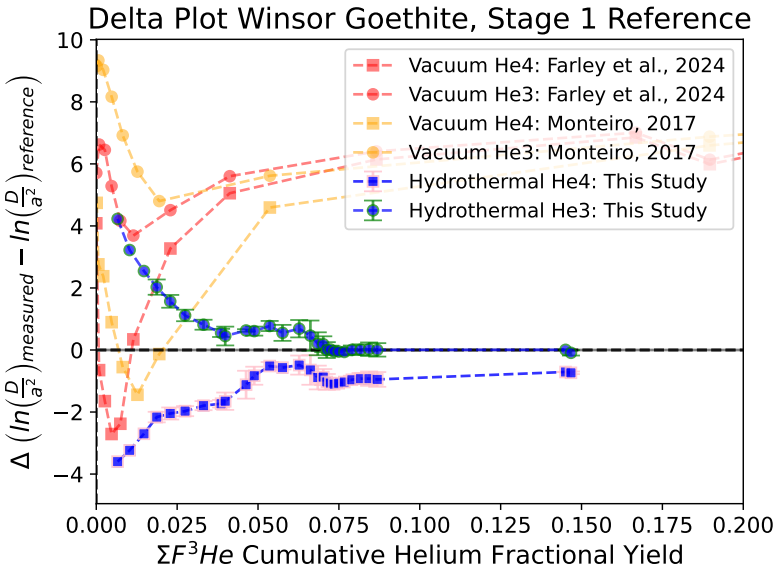


Figure 8: Delta plot after Farley (2018), comparable to the r/r_0 plots of Lovera et al. (1991) and Farley and Flowers (2012). Delta is the difference between apparent log diffusivity and the Stage 1 reference line, which removes the temperature dependence of the dominant migration process to show how apparent diffusivities evolve through the experiment. ^{3}He decreases monotonically, settles on a temporary plateau from 4–6% fractional yield, then follows the reference line from 7–15%. ^{4}He delta values are more uncertain; they increase initially and settle about one log unit below the ^{3}He reference line by about 5% ^{3}He yield.

Bulk-normalized ^{4}He/^{3}He ratios begin at very low values and increase steeply, with the rate of increase slowing as the experiment proceeds (Fig. 9). At any given cumulative ^{3}He yield the hydrothermal experiment records lower bulk-normalized ratios than either vacuum experiment on the same sample, and its approach toward a plateau is the slowest of the three. The hydrothermal experiment did not reach a plateau.

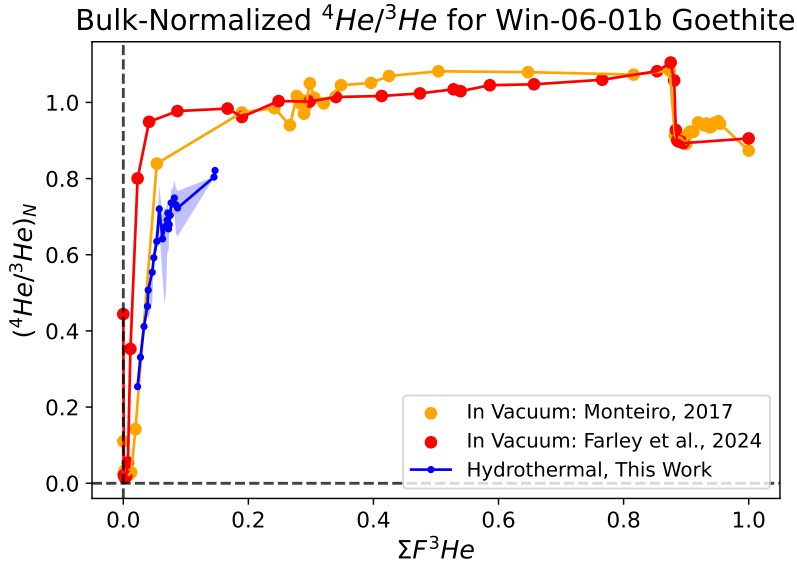


Figure 9: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b step heats against cumulative fractional ^{3}He yield. The hydrothermal experiment records lower ratios at a given yield than either vacuum experiment for the entire duration, apart from the first step. All three begin at very small values and increase steeply, with the rate of increase slowing between 0.05 and 0.2 cumulative yield. The hydrothermal experiment is plotted as an envelope spanning the lower and upper bounds of Eq. (9). Ten of 34 analyses are excluded, either because λt was too small to evaluate bounds or because the measured ^{3}He signal was below 0.1 pccSTP.

5. Discussion

5.1. Encapsulation stabilizes reactive phases during heating

Pyrite, arsenopyrite, and goethite heated without a buffering phase, in vacuum or in open air, develop surface roughening and contraction cracks (Fig. 1, right column). The same phases heated at the same temperatures inside sealed silica glass ampoules containing sulfur, realgar, or water retain the smooth surfaces and clear cleavage faces of the untreated controls (Fig. 1, centre column). For goethite, cracks in the unbuffered material run both parallel and perpendicular to the long axis of the aggregate. Goethite heated in open air turned red; goethite heated in vacuum remained black.

5.2. Recovering diffusivities for encapsulated Durango apatite

Step heats on encapsulated Durango apatite reproduces the diffusion behavior measured on unencapsulated material, following deconvolution of the permeation lag from the sample's own helium release using the protocol described in §3.3. The recovered inverted diffusivities of the encapsulated Durango apatite differ slightly from the reference line taken from Farley (2000) at the very beginning and end of the experiment, consistent with previous work on the sample. Farley (2000) also observed anomalously low apparent diffusivities in early steps. Above 300°C the corrected array curves downward, a feature documented for Durango apatite and attributed to annealing of radiation damage and crystal defects (Wolf, Farley, and Silver, 1996; Farley, 2000; Shuster, Flowers, et al., 2006).

These results seem to be unaffected by several potential complexities:

1. The apatite grains were a substantial portion (≈20%) of the volume of the ampoule. As discussed in §5.3 below, Eq. (10) predicts that this is expected to increase the rate constant of permeation by 25%. Indeed, when we reduce the data using rate constants 25% higher than the literature values, we recover a cleaner arrhenius plot. Complexities such as these further motivate in-situ measurement of the ampoule's rate constant during the experiment as detailed in §3.4. Even still, the ≈25% error in permeabilities does not affect the interpretation of these data.
2. Less retentive phases such as apatite have the potential to produce relatively large concentrations of helium within the glass ampoule. This could dampen the helium concentration gradient between the sample

and the gas phase, slowing helium evolution to produce erroneously low apparent diffusivities when inverted using the equations of Fechtig and Kalbitzer (1966). The ratio of helium concentration in the ampoule gas phase and the sample can be calculated given their relative volumes, the calculated helium inventory of the ampoule, and the remaining helium within the sample. For the Durango experiment, the maximum value of $C^{He}_{gas}/C^{He}_{apatite}$ is between 1 and 2%, which, although unimportant for the scope of this work, may be unacceptable for later studies using this method. For more retentive phases, this is less likely to be a concern: this ratio never exceeded 1‰ for the Win-06-01b step heat experiment.

5.3. Elevated rate constants in water-bearing ampoules

Every measured ^{3}He permeability measured on HT-Win-06-01b equals or exceeds the anhydrous literature value, and the fitted activation energy of 10192 ± 1882 cal/mol is substantially higher than expected for silica glass with or without water (Swets et al., 1961; Jones, 1953; Masaryk, 1968; Shelby, 1972), lying closer to the values reported for glasses rich in network-modifying cations (Norton, 1953; Altemose, 1961). Our measurement of the Win-06-01b helium permeation rate above the anhydrous literature value was, at first, an unexpected result. Water and hydroxyl groups dissolved in silica glass are reported to decrease helium permeability at all temperatures (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968), following the interpretation that species such as OH^- occupy interstitial space in the glass network (Norton, 1953; Shackelford, 1999); However, we have observed anomalously high rate constants in water-bearing ampoules consistently.

The offset is not an artifact of the measurement procedure. Helium continues to permeate while the ampoule cools before each inlet, which inflates the apparent permeability, but the finite-difference simulation of that process predicts an offset that is near constant across the temperature range and smaller than the uncertainties on the fitted values (Fig. 5). It cannot account for a systematic excess over the literature values, nor for the elevated activation energy, which depends on the slope rather than on any constant offset.

The likely explanation is geometric: adding a significant volume of water reduces the volume available to the gas phase without reducing the glass area available for permeation. Helium is only sparingly soluble in water and diffuses through it rapidly, roughly 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984) against about 2×10^{-8} cm^2/s in silica glass (Swets et al.,

1961). On the timescale of permeation through the glass wall, transport through the water is effectively instantaneous, so the gas phase stays in equilibrium with the glass over the whole inner surface, including where liquid water contacts it. Helium partitions strongly into the gas phase over both water and glass: helium concentrations in the glass and water are about 3% and 1% respectively of the gas-phase concentration, using solubilities from Sander (2015) and Swets et al. (1961). Displacing gas-phase volume with water therefore raises the helium partial pressure without changing the surface area available for permeation. Because the rate of helium permeation is proportional to the partial pressure in the gas phase (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), the permeability rate constant is related to the volume fraction of the gas and condensed phases within according to the following relationship, assuming that negligible helium partitions into condensed phases (such as water or the mineral sample),

$$\lambda_{ampoule} = \lambda_{empty} \frac{V_{condensed} + V_{gas}}{V_{gas}}. \tag{10}$$

Where $\lambda_{ampoule}$ is the true rate constant of helium permeation for an ampoule containing a mineral/buffer payload, and λ_{empty} is the rate constant of that same ampoule with only the gas phase inside. This observed complexity emphasizes the need to measure the ampoule's rate constant in-situ during the experiment as described in §3.4.

5.4. Consistency of the bounding method to account for permeation lag for Win-06-01b

Both isotopes were reduced by the bounding route of §3.5 because ^{4}He could not be reduced following the protocol of §3.3 due to blank-level permeation step yields. ^{3}He was also reduced this way for consistency with ^{4}He. As an internal test, the direct quantification of the ampoule's ^{3}He inventory using permeation steps as described in §3.3 is applied in the supplement to demonstrate the validity of the bounding approach. For the bounding method, λ values were conservatively underestimated using geometric measurements of the ampoule and the literature values of ^{4}He permeability in silica glass from Swets et al. (1961). That permeability is then scaled by the ratio of Jones (1953) to get the ^{3}He rate constant. The bounding method produces the 3He Stage 1 regression (§4.3) with activation energies between 157.1–160.6 kJ/mol and closure temperatures between 97.5–101.2°C. Reduc-

ing ^{3}He instead by the direct route of §3.3 using the measured rate constants (§3.3) gives kJ/mol, within uncertainty of the value obtained by bounding.

[direct-route E_a to be inserted]

Fig. 10 shows this comparison directly: n_t, the ampoule's ^{3}He inventory quantified from permeation steps via Eq. (6) (the permeation-step method; supplement), plotted against n^{ss}, the upper bound of Eq. (8) calculated independently from the preceding high-temperature step (the bounding method used throughout this paper). Excepting two steps discussed below, $n_t \leq n^{ss}$ for every permeation step in the experiment, as required if the steady-state approximation genuinely bounds the ampoule's residual helium from above. This is the empirical basis for reducing the ^{4}He dataset, and the ^{3}He dataset for consistency with it, by bounding alone.

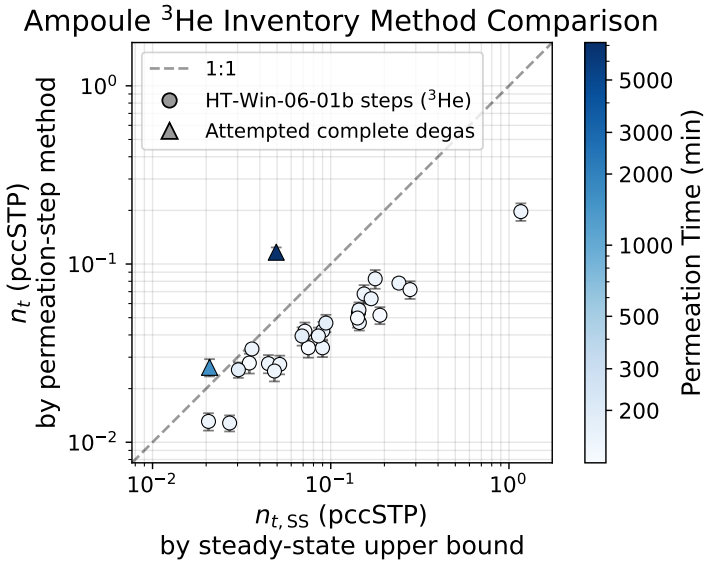


Figure 10: Two independent estimates of the ampoule's residual ^{3}He inventory following each permeation step of HT-Win-06-01b: n_t, quantified directly from the permeation step itself via Eq. (6) (y-axis; supplement), against n^{ss}, the upper bound of Eq. (8) calculated from the preceding high-temperature step (x-axis). Color gives the permeation step duration. Circles are ordinary steps; triangles are the two "attempted complete degas" steps, held at 100°C for far longer than any other permeation step in the experiment (1580 and 7200 minutes). The dashed line is 1:1. Excluding the two complete-degas steps, n_t never exceeds n^{ss}. Both complete-degas steps instead exceed their own n^{ss}.

The two complete-degas steps break this pattern in a way that scales with step duration. The longer step (7200 minutes, for which simple first-order

decay predicts $n_t/n_0 \approx 1 \times 10^{-8}$) exceeds n^{ss} by a factor of 1.5; the shorter step (1580 minutes, $n_t/n_0 \approx 0.02$) exceeds n^{ss} by only 25%. Because Eq. (8) is conservative only if the sample's release rate is constant or decreasing, exceeding it at all means that assumption failed for these two steps: the source term must have increased, not decreased, at some point during the hold.

The excesses also scale with the size of the immediately preceding high-temperature step rather than with anything intrinsic to the ampoule. The 7200-minute step, with the larger excess, followed a high-temperature step that released 8.5 pccSTP of ^{3}He; the 1580-minute step, with the smaller excess, followed one that released only 0.2 pccSTP. A rate-constant error, undetected leak, or extraction-line memory effect would not be expected to track the preceding step's yield from the sample. That the excess does track it points to the goethite rather than the ampoule: whatever is releasing this helium was charged by the preceding high-temperature step, in proportion to its yield, and bled out slowly, and only weakly temperature-dependent, over the following days to weeks at 100°C.

This pattern is harder to reconcile with dehydration than with diffusion. Dehydration of goethite in vacuum step heats elevates apparent helium diffusivity (Farley, H. B. Monteiro, et al., 2024) (§5.6), but the 250°C steps immediately following each complete-degas hold return delta values indistinguishable from zero relative to the Stage 1 regression (Fig. 8), not the elevated delta a newly dehydrated fraction of the aggregate should produce. We prefer instead diffusion of residual helium from interstitial space within the goethite crystallite aggregate (Naono et al., 1987) (§5.5), recharged by the preceding high-temperature step and released slowly during the subsequent low-temperature hold. This requires communication between a lattice or grain-boundary source and an interstitial sink, which the independent, non-communicating domains of a strict multiple diffusion domain treatment do not allow, but it extends rather than contradicts the interstitial framework we already invoke for the low-retention "Stage 1" fraction (§5.5), and unlike dehydration it requires no change in the retentivity of the crystalline lattice recorded by the rest of the experiment.

5.5. Helium release from goethite is consistent with multiple diffusion domain systematics

Natural goethite occurs as aggregates of crystallites typically smaller than 1 µm, and all published goethite step heats are on polycrystalline material

of unclear and probably variable crystallite size (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024). Such a sample should show multiple diffusion domain behavior, and neglecting it produces inaccurate activation energies and frequency factors (Farley, 2018).

In a multiple diffusion domain system with an initially uniform diffusant, apparent bulk diffusivity at fixed temperature decreases as the least retentive domains are exhausted, and delta values decrease correspondingly, provided the reference line's activation energy matches that of the controlling process (Lovera et al., 1991; Farley, 2018). Using the activation energy retrieved here, the ^{3}He delta values decrease monotonically (Fig. 8), which is the expected behavior.

The plateau in the delta plot and the quality of the regression over 7–15% cumulative ^{3}He yield ($R^2 = 0.989$) suggest that a single domain dominates the frequency factor over this interval, rather than a mixture of domains as is typically observed for polycrystalline hematite (Farley, 2018). The ^{4}He apparent diffusivities are lower than the ^{3}He values and increase early in the experiment, which is what rounding of the radiogenic ^{4}He profile by natural loss from the least retentive domains would produce (Shuster, Farley, et al., 2004; Shuster, P. M. Vasconcelos, et al., 2005).

Vacuum experiments on goethite also record declining delta values at low ^{3}He yields that are not mirrored in the ^{4}He data (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). One interpretation is migration out of fast pathways along grain boundaries that would retain little helium in nature (Shuster, P. M. Vasconcelos, et al., 2005). Goethite aggregates contain interstices filled with chemisorbed non-stoichiometric water, which is the first water lost in vacuum (Naono et al., 1987), and non-stoichiometric hydrogen persists in goethite above 140°C (Yapp and Poths, 1995). Proton-induced ^{3}He sited in such interstices may be released with this early water, which would correspond to the low retention domain of Shuster, P. M. Vasconcelos, et al. (2005) and the Stage 1 release of Farley, H. B. Monteiro, et al. (2024). The wide variability in the size of this early fraction is consistent with the strong dependence of goethite crystallinity, surface area, and non-stoichiometric water content on formation temperature (Schwertmann et al., 1985).

These data do not bear on whether helium diffusion in crystalline goethite is anisotropic or modulated by radiation damage (Bassal et al., 2022).

5.6. Dehydration and recrystallization

Bulk goethite is unstable with respect to hematite and water at the pressures and temperatures of this experiment (Majzlan et al., 2003), nanoparticles excepted (Guo and Barnard, 2011; Navrotsky et al., 2008). Dehydration or recrystallization could therefore contribute to the observed release, and neither can be excluded.

The electron microscopy of §5.1 bounds pervasive dehydration. Goethite heated in vacuum or open air develops contraction cracks through grain interiors and across grain surfaces, while goethite heated under water-saturated conditions does not (Fig. 1). This excludes dehydration at the scale that produces visible cracking. It does not exclude dehydration below that scale, and work on the textural evolution of goethite aggregates at low dehydration fractions would be needed to resolve how much can occur before cracks appear.

Recrystallization is harder to bound. Poorly crystalline synthetic goethite recrystallizes under hydrothermal conditions in as little as 16 hours at 125°C (Schwertmann et al., 1985), and our conditions were considerably harsher. The silica dissolved from the ampoule wall and redeposited as a gel-like lining (Fig. 2b) shows that the fluid was capable of dissolving and reprecipitating rock-forming oxides. Whether it recrystallized Win-06-01b specifically is unclear; this sample is among the most crystalline and most helium-retentive goethites yet dated (Waltenberg, 2012; Farley, H. B. Monteiro, et al., 2024), whereas the less crystalline goethites commonly recrystallized in nature appear to lose their helium and reset (H. S. Monteiro, P. M. P. Vasconcelos, et al., 2018; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Heim et al., 2006).

The interpretive consequence is the same either way. If dehydration or recrystallization contributed to the release, the apparent diffusivities reported here are upper bounds on the true helium diffusivity in goethite at these temperatures and yields, and the retentivity we infer is a lower bound. Both would then imply that dehydration and recrystallization, rather than volume diffusion, set the pace of helium loss from goethite in nature.

5.7. Helium retentivity of goethite

Apparent diffusivities for HT-Win-06-01b are resolved from, and about seven natural log units below, those measured for the same sample in vacuum (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024). We consider what

this implies through closure temperature, isothermal retention, and comparison with other polycrystalline samples. Each applies strictly only to loss by thermally activated volume diffusion from domains of known size, so each carries the caveats of §5.6, and none permits estimating the diffusivity at infinite temperature.

Treating the Stage 1 regression as a real diffusion domain gives a closure temperature of 98.1 ± 8.6 °C at 10°C/Ma, considering only the intercept uncertainty. That is the closure temperature of a hematite crystal of about 0.2 µm radius, or an apatite crystal of about 650 µm radius.

Closure temperature is the wrong measure for a mineral that forms at the surface and stays there, since it presumes a cooling rate (Dodson, 1973). Isothermal retention is more appropriate (Wolf, Farley, and Kass, 1998). For a domain with the parameters of the Stage 1 regression held isothermally at 40°C for 100 Ma, $t'/t = 0.968$, where t' is the helium age, t the holding time, and $t'/t = 1$ is perfect retention. At 25°C the same domain is nearly quantitatively retentive at billion-year timescales.

Comparing whole experiments across samples is easier if the heating schedule is collapsed to a dimensionless diffusive timescale, the Fourier number $Fo = Dt/a^2$, calculated against a common reference line. For a single spherical domain the released fraction is a function of Fourier number alone (Crank, 1979; Fechtig and Kalbitzer, 1966), so if the reference activation energy matches that of the controlling process, curves for different samples retain their relative positions regardless of the intercept chosen. In this space, vacuum step heats of goethite and of hematite form well separated populations, with goethite the less retentive, and HT-Win-06-01b is the most retentive polycrystalline sample, at least over the 14.7% of its ^{3}He held in the least retentive domains. This suggests that the helium retentivity of goethite that is not decomposing may exceed that of hematite of comparable size, though experiments on irradiated single-crystal goethite of controlled size would be needed to test it. Vacuum step heats of unrelated goethite samples resemble one another more closely than step heats of the same sample at different water activities.

5.8. Radiogenic helium loss recorded by $^4He/^3He$ ratios

Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples record the rounding of the radiogenic concentration profile by natural loss, provided loss in nature and in the laboratory propagate with the same geometry (Shuster, Farley, et al., 2004; Farley, H. B. Monteiro, et al., 2024). The steep ini-

tial rise and subsequent flattening seen in all three Win-06-01b experiments (Fig. 9) is consistent with complete, then partial, natural loss from the least retentive domains.

The hydrothermal experiment records lower ratios at any given cumulative ^{3}He yield and approaches its plateau most slowly. The three experiments differ systematically in water pressure: Farley, H. B. Monteiro, et al. (2024) heated open to a liquid nitrogen trap, H. Monteiro (2017) at higher water pressures, and this experiment at water saturation. The ordering of the ratio spectra follows that ordering, which suggests the spectra are shaped in part by the extent and spatial propagation of dehydration rather than by natural helium loss alone. The hydrothermal experiment did not reach a plateau and might have with further heating.

Both vacuum experiments show an elevated first-step ratio that is absent here. It represents a negligible fraction of the total ^{4}He and may reflect physisorbed atmospheric helium (Protin et al., 2016), which would have been removed during the four days at 70°C preceding the hydrothermal experiment.

6. Conclusions

Sealing a sample in a pure silica glass ampoule preserves reactive phases through helium step heating and allows the experiment to be run under controlled water or sulfur fugacity in a standard ultra-high-vacuum heating cell, at internal water pressures up to about 86 bar at 300°C. Helium permeates the ampoule wall as a first-order process, and the lag this introduces between release and measurement can be removed either by measuring the ampoule's residual inventory with low-temperature permeation steps or, for long steps, by bounding it. Encapsulated Durango apatite recovers its published Arrhenius array under this correction. The ampoule's permeation rate constant can be measured during the experiment itself, which is necessary for hydrothermal work because water-bearing ampoules do not follow published values for anhydrous glass.

The first hydrothermal helium step heat of goethite gives apparent diffusivities approximately seven natural log units below those obtained from the same sample in vacuum. The corrected ^{3}He array is linear above 7% cumulative yield, with an activation energy of 157.1 $\pm$ 5.3 kJ/mol, and its delta plot follows multiple diffusion domain systematics. Extrapolated to surface

conditions, a domain with these parameters is nearly quantitatively retentive of helium at billion-year timescales at 25°C.

The immediate consequence is for how vacuum step-heat data on goethite should be read. The difference between the two experiments on the same aliquot of the same sample is far larger than the differences between vacuum experiments on unrelated samples, which indicates that vacuum step heats of goethite are dominated by the response of the mineral to the experiment rather than by helium migration through it. Helium retention in natural goethite is correspondingly more robust than vacuum experiments imply.

We cannot exclude a contribution from dehydration or recrystallization under hydrothermal conditions. If either contributed, the diffusivities reported here are upper bounds and the retentivity a lower bound, which would strengthen rather than weaken the conclusion. Distinguishing these possibilities, and separating the effects of crystallite size, chemical composition, and radiation damage, will require hydrothermal experiments on irradiated single-crystal goethite of controlled size. The method described here makes those experiments straightforward.

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